

Assessment of Duisport's Environmental Impact on the City of Duisburg

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Contents

Abstract.....	- 3 -
1. Introduction	- 3 -
1.1. Background Information	- 3 -
1.2. Motivation.....	- 4 -
2. Comparing the Environmental Impact of Different Means of Transportation.....	- 5 -
3. The Environmental Impact of Cargo in Duisburg.....	- 6 -
3.1. What the Duisburger Hafen AG Is Doing for the Environment	- 6 -
3.2. Suggestions for the Duisburger Hafen AG	- 8 -
4. Problems	- 8 -
5. Conclusion	- 9 -
References.....	- 10 -

Abstract

With the expansion of the harbour of Duisburg, the debate over the quality of life in the neighbouring homes becomes more relevant. This paper aims to assess the environmental impact of the harbour of the Duisburger Hafen AG (called duisport). There will be a closer look at the specifics of the new Silk Road of the Belt and Road initiative and its impact. This is achieved by combining the results of different research which will be projected on the duisport. This will be opposed by the green projects, that the duisport has completed or raised interest in. This paper concludes that the duisport has a negative impact on the surrounding city, but duisport is investing in greener solutions and is trying to compensate for their negative impact. There could be done more for a greener environment, but this always must be feasible.

Keywords: Duisport, Belt and Road Initiative, environmental impact, New Silk Road

1. Introduction

The duisport is Europe's biggest inland harbour. In 2020 the harbour of duisport had a cargo turnover of 59 million metric tons (Duisburger Hafen AG, 2021a, p. 1). With the duisport being the end of the new Silk Road and a large end stop for the maritime Silk Road, comes more freight to the harbour. This may indicate a negative impact on the quality of life in Duisburg, because of air pollution. This study aims to assess the environmental impact of the harbour on the city of Duisburg.

1.1. Background Information

The domestic coal production slowly faded out in the previous years. Due to the "oil price shock" in the late 1970s, importing oil became so expensive that domestic coal momentarily appeared to be viable again. This led to a "cycle of crisis" in the coal and steel sector (Dahlbeck et al., 2022, p. 57). According to a newspaper article of the Rheinische Post (2009), due to this event, Thyssen AG had to cut around 8,000 jobs from the total of 58,000 in 1993. Of those 7,000 were cut in Duisburg. Thyssen AG is, now under the name Thyssenkrupp AG, a large steel manufacturer in the area. To open different

possibilities other than steel, the city of Duisburg signed up for a Freeport. A Freeport is a port with no import regulations. This was the first step toward a better import/export-evolution (Rexing, 2020, p. 1). Today the harbour of Duisburg is operated by the Duisburger Hafen AG and is called the duisport. Today the harbour is more than just an inland harbour, it is also a large train port and loading and unloading station in Europe. The harbour is located ideally for the western end of the new Silk Road (Pascha, 2021, p. 19). The duisport calls itself the “gateway to Europe” as they claim to reach 30 million customers and 300.000 companies, with a total purchasing power of 600 billion euros of purchasing power within 150 kilometres of the harbour (Duisburger Hafen AG, 2021b). From here all the goods from Eurasian trains are distributed throughout the whole of Europe. The Belt and Road Initiative (BRI) was initiated by the Chinese government in 2013 and aims to connect Asia to Europe and Africa (Belt and Road Initiative, 2022). This is mostly achieved by investments in infrastructure, for example, the new Silk Road and the maritime Silk Road. The new Silk Road is a railway network connecting multiple states and train ports between Europe, Asia, and Africa. The maritime Silk Road is a seaway network connecting many states primarily those in Asia and Africa. The influence of the duisport on its surrounding environment and the quality of life in the city seems to be negative. To evaluate the environmental impact, we first must look at the environmental impact of the individual transportation methods and then the cargo turnover of the harbour. After that, we can estimate the impact of the harbour on the city.

1.2. Motivation

This essay will consider the environmental impact of the duisport on Duisburg only as the impact of the cargo entering and exiting the ports of the Duisburger Hafen AG by truck, ship, and train. This scope was chosen for this paper because the new Silk Road ends in Duisburg, on the European side. This means a lot of trains arrive here which are packed with cargo that must be distributed to other states in Europe. So not only the traffic for the import and export of Germany is processed in Duisburg but for many other countries as well. This could indicate a bigger negative impact on the city,

due to the larger import and export of goods. Measuring this is quite difficult and due to the scope of this paper, this was simplified quite a bit. The problems, concerning the simplification will be addressed later in this paper.

2. Comparing the Environmental Impact of Different Means of Transportation

In this section, this paper will firstly look at the greenhouse gas emission differences between the different transportation methods. Secondly, this section will be looking at the fact that ships must take water routes and, as a result, most must travel way longer distances which takes more time.

Long-distance cargo transportation by train has a higher greenhouse gas emission than by ship, but less than by truck or by plane. According to Mathers et al. (2019), the greenhouse gas emission per one-ton mile is about 1.6 times higher for freight trains than for cargo ships. A one-ton mile is the equivalent to transporting one ton of products, one mile. Trucks have 10 times bigger emissions than cargo ships and aeroplanes have 47 times larger emissions Mathers et al., 2019, p. 27). Though Mathers et al. (2019) refer to train transportation powered by diesel. When using electric trains, the greenhouse gas emission of the train during transport would be zero. A study by Zhang et al. (2014) showed, that the abiotic depletion potential (ADP) of electric locomotives is significantly lower than the ADP of diesel locomotives (Zhang et al., 2014, p. 150). The ADP measures resource consumption, meaning the not returnable resources consumed. This can happen by just straight-up losing resources but mostly contaminating resources so they cannot be used again (Dincer & Abu-Rayash, 2020, p. 145).

Ships must use sea routes which are mostly longer than the land equivalent. A great example of this is the difference between the new Silk Road and the maritime Silk Road, also called the Suez Route, because it passes through the Suez Canal. The new Silk Road has a length of around 10,000 kilometres and a train from Asia to Duisburg needs around 12 to 14 Days (DB Cargo, 2021). A ship needs to drive around 22,000 kilometres and takes approximately double the time (DSV A/S, 2022). This raises the question, of why more cargo is still being delivered by ship. The short answer

to this question is the price. Sending cargo via Ship takes longer and is worse for the environment but quite a lot cheaper. According to DSV A/S (2022), a Danish transport and logistics company, transporting 22,000 kilograms of goods, equal to one 40-foot container, cost around 8,000 US dollars by train, 4,000 US dollars by sea and 32,000 US dollars by air for the distance between China and Europe.

3. The Environmental Impact of Cargo in Duisburg

The previous sections showed that the greenest option is cargo by electric freight trains. However, those cannot be deployed on most track sections as they do not feature catenaries. The previously shown, most environmentally friendly alternative, is the ship. But the annual report of the Duisburger Hafen AG for the year 2020 showed, that most cargo is still being delivered by truck, which, as shown before, is quite bad for the environment (Duisburger Hafen AG, 2021a). This has a direct impact on the City of Duisburg, because of the high percentage of cargo by truck. Trucks not only produce a lot of greenhouse gases but also noise pollution. Not all pollution is emitted inside the city, but this city was chosen because many trucks deliver cargo to and from the harbour. The port itself must be prepared to accommodate a greener solution, for example electric trains.

But with all this information we still must consider the history of the city. The Duisburger Hafen AG bought old brownfield sites in 1998 to expand (Duisburger Hafen AG, 2022b, p. 1). It could be concluded, that more cargo results in more environmental damage, due to assumed future growth of the company (Duisburger Hafen AG, p. 1).

3.1. What the Duisburger Hafen AG Is Doing for the Environment

This section will introduce three projects, the first one being a hydrogen-powered shunting locomotive, the second one being the “enerPort II” and the third is called MagRail. The goal of all three projects is to decrease the negative impact on the environment. Especially considering that the duisport

is growing significantly, thus the volume of greenhouse gases emitted by the harbour is expected to increase accordingly. The Duisburger Hafen AG is building two new so-called “logports” with a total space requirement of 65 hectares. “logport” is the project name for Duisport’s logistics centres. (Duisburger Hafen AG, 2022b).

As stated in a preliminary study on hydrogen-powered locomotives for the Duisport, there is a need for renewal, but a hydrogen-powered locomotive is not economically smart when the train tracks feature catenary along the whole railway (Konrad et al., 2022). The cheaper and more environmentally friendly way would be an electric locomotive. Also, diesel and electric-powered locomotives are cheaper per Megajoule than hydrogen-powered ones. If the train tracks do not feature a catenary, the diesel locomotive would still be cheaper but worse for the environment (Zhang et al., 2014, p. 150).

The business partner of the Duisburger Hafen AG for the “enerPort” is the Fraunhofer UMSICHT. They point out, that concerning the energy transition, inland ports are also interesting urban areas with their own requirements profile and great development potential. In addition to the Duisport, other industrial companies are located in the ports, and due to the proximity to residential areas, inland ports are increasingly responsible for contributing to an attractive living environment. (Fraunhofer-Institut für Umwelt-, Sicherheits- und Energietechnik UMSICHT, 2019). The Duisburg Gateway Terminal (DGT) will incorporate the “enerPort II”, which will implement a sustainable, hydrogen-using energy concept that aims for a high degree of self-sufficiency (Fraunhofer-Institut für Umwelt-, Sicherheits- und Energietechnik UMSICHT, 2022). As stated in the press release on the third of May 2022 the working processes of the DGT will be 100 per cent environmentally friendly. The DGT is considered the leading example of the energy transition for inland harbours (Duisburger Hafen AG, 2022a). Many different inland harbours already raised interest in the “enerPort” concept (Berger et al., 2021, p. 55).

Another interesting and recent project is the MagRail. Nevomo, the company behind MagRail, announced on the fifth of May 2022 that they and

the duisport “[...] signed a Memorandum of Understanding (MoU) to explore the possibility of expanding duisport's capacity with MagRail within the port's existing infrastructure.” (Nevomo, 2022, p. 1)

As stated in the climate protection concept for the city of Duisburg, the Integrated Energy & Climate Protection Concept of Duisburger Hafen AG (IEKK duisport) was developed jointly with the Centre for Logistics and Transport (ZLV) at the University of Duisburg-Essen. Sustainability and climate protection in particular have been an integral part of duisport's business areas, projects and activities for many years and are often part of the core of the business (Hoene et al., 2017, p. 178).

3.2. Suggestions for the Duisburger Hafen AG

A recent study proposed that battery-based last-mile modules for freight locomotives would make catenaries at the railway station obsolete (Brenna et al., 2019, p. 1). This is especially interesting for the duisport because the duisport does not feature a catenary above most tracks. Another option would be investing in hydrogen-powered, catenary-powered, battery-powered locomotives or the previously mentioned MagRail because all these options would remove any direct emission to the city.

4. Problems

There are many problems when trying to estimate the environmental impact of duisport. This paper only and specifically looked at the impact of cargo ships, freight trains, and trucks. It also shortly revised the impact of shunting the cargo at the port. Evaluating every aspect of the duisport would reach beyond the scope of this paper. The factors influencing the actual environmental impact of the duisport are far more complex and include, for example, the investments and research done by the Duisburger Hafen AG on renewable energy resources. They state that they take the environmental aspect very serious (Duisburger Hafen AG, 2019). It's unsure on which level the duisport is just using this for public relations. This paper estimated the environmental impact differences between the different transportation methods on a broader example and then projected these findings on the duisport and linked these to the cargo turnover of the Duisburger Hafen AG.

These abstractions lead to a noise in the projected findings. More research could be done by conducting studies about the energy consumption of the Duisport and the types of energy used.

5. Conclusion

In conclusion, the BRI and the cargo being delivered by freight trains, cargo ships, and trucks do harm the environment. The differences between the different delivery methods are larger than expected and cargo ships are still the eco-friendliest option. Only using eco-friendly shunting locomotives would be a good way to decrease the direct air pollution inside the city. Due to the expansion of the Duisport, an attractive living environment becomes more and more important for the city. This study has also elaborated, that the Duisburger Hafen AG is taking this seriously and is investing in greener solutions, the effectiveness remains to be seen.

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